

OMU, Japan

WMO: 474050

Lat: 44.5806N

Lon: 142.9636E

Elev: 52

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.8	6.6	-1.9	5.0	3.9	1.4	5.9	7.8	24.7	26.3	21.5	26.0	6.8	230	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	81.9	70.1	78.2	68.2	74.8	66.8	72.6	79.2	70.5	75.8	68.6	73.0	11.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	111.4	76.7	68.5	105.1	74.3	66.7	98.5	72.0	36.4	79.5	34.6	75.8	32.9	73.3	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	19.1	16.4	-3.7	89.0	4.1	3.4	-6.7	91.4	-9.1	93.4	-11.4	95.3	-14.3	97.8
			-4.2	76.1	4.0	2.0	-7.1	77.5	-9.4	78.7	-11.7	79.8	-14.6	81.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	19.8	19.6	28.0	39.0	47.8	54.0	61.8	66.1	60.8	49.4	36.7	25.2	(6)
	DBStd	17.33	6.21	7.38	7.25	7.20	8.35	6.80	6.54	5.68	5.44	5.94	7.45	6.36	(7)
	HDD50	4240	936	850	683	342	145	29	0	0	0	84	403	768	(8)
	HDD65	8385	1401	1270	1147	780	536	338	145	53	149	485	848	1233	(9)
	CDD50	1498	0	0	0	12	77	149	366	501	325	64	4	0	(10)
	CDD65	170	0	0	0	0	3	8	46	89	24	0	0	0	(11)
	CDH74	902	0	0	0	2	47	72	261	427	93	0	0	0	(12)
	CDH80	187	0	0	0	0	8	13	62	94	10	0	0	0	(13)
(14) Wind	WSAvg	7.1	7.8	7.5	8.1	8.2	7.1	5.5	5.0	5.5	6.9	7.9	7.9	7.9	(14)
(15) Precipitation	PrecAvg	37.6	2.0	1.6	1.7	2.0	2.5	2.7	4.6	5.1	5.6	3.5	3.5	2.9	(15)
	PrecMax	51.8	4.1	3.5	3.3	5.1	6.0	6.9	11.1	13.4	13.5	5.5	6.9	5.4	(16)
	PrecMin	24.3	1.0	0.6	0.3	0.2	0.8	1.1	0.7	1.8	2.5	1.1	1.7	1.6	(17)
	PrecStd	6.3	0.7	0.7	0.7	1.0	1.3	1.3	2.5	3.1	3.0	1.3	1.5	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.4	44.0	51.6	69.5	79.9	81.4	86.3	87.1	80.6	70.0	61.8	46.8	(19)
		MCWB	34.0	37.4	42.7	52.7	59.5	64.9	73.7	74.1	68.2	59.0	51.6	41.4	(20)
	2%	DB	34.0	38.5	45.9	61.7	72.6	74.9	80.8	82.4	76.5	65.9	56.2	41.5	(21)
		MCWB	31.2	33.3	38.8	48.9	56.4	61.8	71.2	71.5	66.4	56.7	48.5	37.5	(22)
	5%	DB	31.8	34.5	42.0	56.3	66.9	69.2	76.3	78.8	73.3	63.0	52.4	37.8	(23)
		MCWB	29.8	30.4	36.1	45.9	54.4	59.9	69.0	70.3	65.2	55.1	46.0	34.4	(24)
	10%	DB	30.0	31.3	38.5	51.7	61.6	64.8	71.8	75.2	70.5	60.2	48.9	35.1	(25)
		MCWB	27.7	28.4	33.6	43.4	52.2	58.8	66.7	68.8	63.6	53.6	44.2	32.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	37.6	43.2	54.1	61.6	66.5	75.4	75.6	71.5	61.3	53.1	42.2	(27)
		MCDB	37.3	42.8	50.6	65.7	74.4	77.1	83.5	84.9	77.5	66.9	60.2	45.6	(28)
	2%	WB	31.9	33.5	39.3	49.7	58.1	63.5	72.3	73.2	68.8	58.5	49.3	38.1	(29)
		MCDB	33.5	37.8	45.0	60.3	69.6	72.1	79.1	80.2	74.0	63.5	54.6	40.9	(30)
	5%	WB	29.8	30.7	36.4	46.6	55.6	61.2	69.6	71.3	66.5	56.5	46.7	34.8	(31)
		MCDB	31.6	33.7	41.3	55.0	65.4	67.3	74.8	76.7	71.0	61.4	51.4	37.4	(32)
	10%	WB	27.8	28.7	34.0	43.8	52.9	59.0	66.9	69.7	64.7	54.4	44.3	32.4	(33)
		MCDB	29.8	31.2	37.9	51.1	60.0	64.0	71.2	74.1	69.0	59.2	48.7	34.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	15.8	13.8	15.0	15.3	12.5	10.6	11.6	15.4	16.4	13.1	12.5	(35)
		MCDBR	14.2	17.8	18.0	24.0	25.9	21.9	18.6	18.2	18.7	19.7	19.2	14.8	(36)
	5% WB	MCWBR	12.4	14.3	12.7	14.1	13.3	11.7	10.1	9.3	10.9	12.5	13.5	12.2	(37)
		MCWBR	13.3	16.3	17.3	22.4	23.1	19.1	16.9	15.1	15.4	17.7	17.7	14.1	(38)
(40) Clear-Sky Solar Irradiance	taub		0.339	0.386	0.451	0.513	0.529	0.508	0.487	0.461	0.409	0.407	0.385	0.357	(40)
	taud		2.028	1.847	1.783	1.805	1.868	2.018	2.146	2.239	2.329	2.173	2.126	2.042	(41)
	Ebn at Noon		231	239	241	239	240	246	250	252	257	235	214	208	(42)
	Edh at Noon		34	49	60	64	62	54	47	41	35	36	31	31	(43)
(44) All-Sky Solar Radiation	RadAvg		397	679	1052	1396	1531	1537	1438	1329	1128	782	431	321	(44)
	RadStd		42	94	97	119	124	162	182	87	96	46	44	27	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page